Math **163** Name (PLEASE PRINT) Srinarayan Srikanth

Exam **1C** **2026** Fall/Spring/Summer I /**Summer II** Score 0

Please fill in the following table putting in expected points corresponding to the parts of problems that you have completed and submitted and an expected total. Leave blank those parts that you have not completed or that do not exist. You will lose double the points that a part is worth if you submit incomplete, unproved, or unchecked answers, or if you do not follow the directions of the problem. Ideally you should know exactly what you have done correctly and how many points you should receive for parts and in total. I will fill in the received grade after I correct the exam. Not all questions have the same number of parts so that some columns may be redundant for particular problems.

										Expected	Received
	Part	Part	Part	Part	Part	Part	Part	Part	Part	Grade	Grade
	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)		
Number											
1a											
1b											
1c											
2a											
2b											
2c											
3a											
3b											
3c											
								Total			

Instructions

- All work must be done using a **MECHANICAL PENCIL following instructions given**.
- If you have any questions at all, please **ask me**. This includes the following possibilities:
    - something is **illegible,** or **ambiguous**, or **unclear** to you.
- **You may use your homework, notes, books, and calculators unless otherwise stated during the exam.**
- **You will not get any partial credit**. You have to show **absolutely all your work** on all questions, **exactly as in the notes, with no changes, variations, or omissions in form, unless otherwise noted whether orally or in writing**. If your answer is correct, but your procedure is incorrect or is not allowed by the instructions, you will get no credit at all. All work must be displayed using **correct syntax**, and all **explanations**, written in **complete sentences**, must be complete.
- **You must write neatly, clearly, and legibly. You must draw every straight-line segment other than those in the list: (**$=, \neq, \equiv, <, \leq, >, \geq, +, -, \times, \div, \ \neg, \vee, \wedge, \parallel, \perp, \setminus, \rightarrow, \leftrightarrow, \mapsto$**) with your ruler. At least 1 point will be taken off for each instance of violation. Signs of equality or equivalence must be aligned vertically, one below the other, on the left side of expressions. Every symbol has a specific meaning and may be used only in this sense and in no other. Syntax, defined in the notes, must be followed exactly to the letter.**
- Remember that 'does not make sense' is a possible answer.
- I can grade you only on what you **actually** put on paper; I cannot read minds. You cannot say later that what you had written down is not what you had meant. **You need to learn to write exactly what you mean and mean exactly what you write**.
- **The answers for a problem should be noted on the page on which the problem appears in the areas indicated. All work for the problem should be shown on sheets that follow the page on which the problem appears numbered as follows. Work for a problem that occurs on page n should be on blank (not ruled) sheets after page n, numbered, n0, n1, n2 etc. on the lower right-hand corner as in this document. For example, work for a problem on page 502 should be on pages numbered 5020, 5021, 5022 etc. as needed. Work that corresponds to different parts must be separated from each other by straight lines drawn with a ruler parallel to the edges of the paper and labelled correctly as in the Pages from Student Work as found under Files on CANVAS.**
- **You should submit a SINGLE document.**
- **You may NOT include pages that you have NOT attempted**.
- **As shown in the class means as shown in the notes.**

Instructions:

Use these instructions for the remainder of the exam.

Your J-number

$$JN := (J_k \in 0..9 | k \in 1..8) = \underline{\quad 00499317 \quad}$$

Hence

$$J_1 = 0 \quad J_2 = 0 \quad J_3 = 4 \quad J_4 = 9 \quad J_5 = 9 \quad J_6 = 3 \quad J_7 = 1 \quad J_8 = 7$$

R Write **the elements of** every numerical set or multiset appearing in this exam **in increasing order.**

D $\mathcal{U} := 0..9 = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$

D $SJ := St(J) := \left\{ J_k \in 0..9 \middle| k \in 1..8 \right\} = \{0, 1, 3, 4, 7, 9\}$

$$\nu(SJ) = 6$$

D $Lst(J) := \left((SJ)_k \in 0..9 \middle| k \in 1..(\nu(SJ)) \right) = (0, 1, 3, 4, 7, 9)$

D $\forall l \in 1..(\nu(SJ)) \quad n_l := \nu\left(\left\{ k \in 1..8 \middle| J_k = (SJ)_l \right\} \right)$

D $MSJ := MltSt(J) := \left\{ (SJ)_m : n_m \middle| m \in 1..(\nu(SJ)) \right\}$

$$= \{0:2, 1:1, 3:1, 4:1, 7:1, 9:2\}$$

D $v(MSJ) = \sum_{m \in 1..(v(SJ))} n_m = 8$

D $s := mxm\{k \in 1..8 | J_k = 0\} = 2$

D $b := mnm\{J_k \in 0..9 | Evn(J_k)\} = 0$

D Define Y : = Yes N : = No Pf : = Proof W : = Witness

D $\forall x \in \mathbb{Z}$ $Evn(x)$ $:\Leftrightarrow$ x is even.

D $\forall x \in \mathbb{Z}$ $Odd(x)$ $:\Leftrightarrow$ x is odd.

D $\forall x \in \mathbb{N}$ $Prm(x)$ $:\Leftrightarrow$ x is prime.

R **Note that in a set an element may occur exactly once**. Therefore, there may not be any repeated elements in any set in the above. **So, if you do not have the correct sets, all your answers will be wrong.**

R If answers to different instances of some question on the exam contradict each other you will get a score of zero for every instance.

R **You will lose points for making any extraneous mark on the exam.**

R You may only submit complete answers to questions marked explicitly on the exam. You will lose points for unfinished work.

R Separate pieces of work for any part of any problem recorded on a blank page intended for the purpose must be indexed by the index of the problem and also by numbers starting at 0 that record the order in which the pieces were completed; such parts must be separated from each other by straight lines parallel to the edges of the paper, drawn with a ruler. You will lose points otherwise.

R Any sloppiness, untidiness, and any departure from proper format (as indicated in class) will lead to a score of 0.

6 (1a) Compute, $\forall n \in \mathbb{N}, \forall k \in 0..n$ the following, and express the result as a polynomial in n. **Show computations or substitutions into appropriate formulas on pages that follow, numbered as per instructions, and write the answers on this page. You must use your own numbers before computing.**

2 (i) $$C\left(C\left(v\left(St\left(J\right)\right), v\left(St\left(J\right)\right)\right), P\left(v\left(MltSt\left(J\right)\right), v\left(MltSt\left(J\right)\right)\right)\right)$$

$= \binom{1}{(8!)}$ does not make sense

2 (ii) $$C\left(P\left(v\left(MltSt\left(J\right)\right), v\left(St\left(J\right)\right)\right), P\left(v\left(St\left(J\right)\right), v\left(St\left(J\right)\right)\right)\right)$$

$$= \binom{\frac{(8!)}{(2!)}}{(6!)}$$

2 (iii) $$C\left(P\left(v\left(MltSt\left(J\right)\right), v\left(MltSt\left(J\right)\right)\right), P\left(v\left(St\left(J\right)\right), v\left(St\left(J\right)\right)\right)\right)$$

$$= \binom{(8!)}{(6!)}$$

1a.i.0

$$\frac{C(n,n) = 1}{C(v(St(J)),v(St(J))) = 1} \left\langle n \longleftarrow v(St(J)) \right\rangle$$

1a.i.1

$$\frac{P(n,n) = n!}{P(v(MltSt(J)),v(MltSt(J))) = (v(MltSt(J)))!} \left\langle n \longleftarrow v(MltSt(J)) \right\rangle$$

1a.i.2

$$\begin{aligned} & C(C(v(St(J)),v(St(J))), P(v(MltSt(J)),v(MltSt(J)))) \\ = \; & C(1, ((v(MltSt(J)))!)) \\ = \; & C(1, 8!) \\ = \; & \binom{1}{(8!)} \end{aligned}$$

1a.i.3

$$\frac{\forall n \in \mathbb{N}\ \forall k \in 0..n\ \binom{n}{k} \qquad (8!) \notin 0..1}{\binom{1}{(8!)} \text{ does not make sense}}$$

1a.ii.0

$$\frac{P(n,k) = \frac{(n!)}{((n-k)!)}}{P(v(MltSt(J)), v(St(J))) = \frac{(v(MltSt(J)))!}{((v(MltSt(J)))-(v(St(J))))!}} \left\langle \begin{matrix} n \longleftarrow v(MltSt(J)) \\ k \longleftarrow v(St(J)) \end{matrix} \right\rangle$$

1a.ii.1

$$\frac{P(n,n) = n!}{P(v(St(J)),v(St(J))) = (v(St(J)))!} \left\langle n \longleftarrow v(St(J)) \right\rangle$$

1a.ii.2

$$\begin{aligned} & C(P(v(MltSt(J)), v(St(J))), P(v(St(J)), v(St(J)))) \\ = \; & C\left(\frac{(v(MltSt(J)))!}{((v(MltSt(J)))-(v(St(J))))!}, (v(St(J)))!\right) \\ = \; & C\left(\frac{(8!)}{((8-6)!)}, (6!)\right) \\ = \; & C\left(\frac{(8!)}{(2!)}, (6!)\right) \\ = \; & \left(\frac{\frac{(8!)}{(2!)}}{(6!)}\right) \end{aligned}$$

1a.iii.0

$$\frac{P(n,n) = n!}{P(\nu(MltSt(J)),\nu(MltSt(J))) = (\nu(MltSt(J)))!} \left\langle n \longleftarrow \nu(MltSt(J)) \right\rangle$$

1a.iii.1

$$\frac{C(n,n) = 1}{C(\nu(St(J)),\nu(St(J))) = 1} \left\langle n \longleftarrow \nu(St(J)) \right\rangle$$

1a.iii.2

$$\begin{aligned}
& C(P(\nu(MltSt(J)),\nu(MltSt(J))), P(\nu(St(J)),\nu(St(J)))) \\
= \; & C((\nu(MltSt(J)))!, (\nu(St(J)))!) \\
= \; & C((8!), (6!)) \\
= \; & \begin{pmatrix} (8!) \\ (6!) \end{pmatrix}
\end{aligned}$$

6 (1b) P**rovide complete reasoning by making a systematic, exhaustive list in the form of a table,** to generate, list, and count the following, **showing all work on pages that follow, numbered as per instructions, and writing only the answers on this page**:

R The symbol ∤ means does not divide.

3 (i) D $F \coloneqq \left\{ (x,y) \in \left(St\left(J \right) \right) \times \left(St\left(J \right) \right) \middle| x \nmid y \right\}$

compute: $F =$ {(0,1), (0,3), (0,4), (0,7), (0,9), (3,1), (3,4), (3,7), (4,1), (4,3), (4,7), (4,9), (7,1), (7,3), (7,4), (7,9), (9,1), (9,3), (9,4), (9,7)}

compute: $\nu\left(F \right) =$ 20

3 (ii) D $G \coloneqq \left\{ (x,y) \in \left(St\left(J \right) \right) \times \left(St\left(J \right) \right) \middle| y \nmid x \right\}$

compute: $G =$ {(1,0), (1,3), (1,4), (1,7), (1,9), (3,0), (3,4), (3,7), (3,9), (4,0), (4,3), (4,7), (4,9), (7,0), (7,3), (7,4), (7,9), (9,0), (9,4), (9,7)}

compute: $\nu\left(G \right) =$ 20

1b.i

$F \coloneqq \{(x,y) \in (St(J))\times(St(J)) \mid x \nmid y\}$

(x,y)	0	1	3	4	7	9
0	(0,0)	(0,1)	(0,3)	(0,4)	(0,7)	(0,9)
1	(1,0)	(1,1)	(1,3)	(1,4)	(1,7)	(1,9)
3	(3,0)	(3,1)	(3,3)	(3,4)	(3,7)	(3,9)
4	(4,0)	(4,1)	(4,3)	(4,4)	(4,7)	(4,9)
7	(7,0)	(7,1)	(7,3)	(7,4)	(7,7)	(7,9)
9	(9,0)	(9,1)	(9,3)	(9,4)	(9,7)	(9,9)

$$F = \left\{ \begin{matrix} (0,1), (0,3), (0,4), (0,7), (0,9), (3,1), (3,4), (3,7), (4,1), (4,3), \\ (4,7), (4,9), (7,1), (7,3), (7,4), (7,9), (9,1), (9,3), (9,4), (9,7) \end{matrix} \right\}$$

1b.ii

$G \coloneqq \{(x,y) \in (St(J))\times(St(J)) \mid y \nmid x\}$

(x,y)	0	1	3	4	7	9
0	(0,0)	(0,1)	(0,3)	(0,4)	(0,7)	(0,9)
1	(1,0)	(1,1)	(1,3)	(1,4)	(1,7)	(1,9)
3	(3,0)	(3,1)	(3,3)	(3,4)	(3,7)	(3,9)
4	(4,0)	(4,1)	(4,3)	(4,4)	(4,7)	(4,9)
7	(7,0)	(7,1)	(7,3)	(7,4)	(7,7)	(7,9)
9	(9,0)	(9,1)	(9,3)	(9,4)	(9,7)	(9,9)

$$G = \left\{ \begin{matrix} (1,0), (1,3), (1,4), (1,7), (1,9), (3,0), (3,4), (3,7), (3,9), (4,0), \\ (4,3), (4,7), (4,9), (7,0), (7,3), (7,4), (7,9), (9,0), (9,4), (9,7) \end{matrix} \right\}$$

5 (1c) We define: $\rho \coloneqq n \longmapsto 2n(n+1) : \mathbb{N} \longrightarrow \mathbb{N}$

Prove without using induction that: $\forall n \in \mathbb{N}\left(4|\left(\rho(n)\right)\right)$.

Fill in the blank: $\forall n \in \mathbb{N} \quad \rho(n) \coloneqq 2n(n+1)$

Begin your proof on this page and continue onto pages that follow, numbered as per instructions.

SPf

$$\frac{n\in\mathbb{N}}{Evn(n) \vee Odd(n)}$$

$$\left(\begin{array}{c} \dfrac{\dfrac{Evn(n)}{n=2m}}{\begin{array}{l}\rho(n)\\ =\ 2n(n+1)\\ =\ 2(2m)(n+1)\\ =\ (m(n+1))(4)\end{array}} \\ \hline 4|\rho(n) \end{array}\right)$$

$$\left(\begin{array}{c} \dfrac{\dfrac{Odd(n)}{n+1=2k}}{\begin{array}{l}\rho(n)\\ =\ 2n(n+1)\\ =\ 2n(2k)\\ =\ (nk)(4)\end{array}} \\ \hline 4|\rho(n) \end{array}\right)$$

$$\frac{Evn(n) \vee Odd(n)}{4|\rho(n)}$$

$\forall n\in\mathbb{N}(4|\rho(n))$ EPf

6 (2a) **Use a tree** (**which you should draw on pages that follow, numbered as per instructions**) to **generate**, **list**, and **count** the elements of the set of all **3-letter words** in the alphabet $\{s, b\}$ such that

My $\{s, b\} = \{ \quad 2, 0 \quad \}$

3 (i) order matters and repetition is allowed, and

$$Wrd\left(3, \{ \quad 2, 0 \quad \}, OM, RA\right) = \left\{ \begin{array}{l} 000, 002, 020, 022 \\ 200, 202, 220, 222 \end{array} \right\}$$

$$\nu\left(Wrd\left(3, \{ \quad 2,,0 \quad \}, OM, RA\right)\right) = 8$$

3 (ii) order does **not** matter and repetition is allowed

$$Wrd\left(3, \{ \quad 2,,0 \quad \}, \cancel{OM}, RA\right) = \{ 000, 002, 022, 222 \}$$

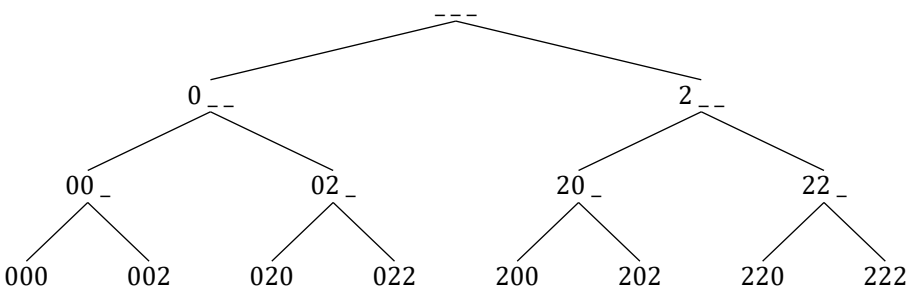
$$\nu\left(Wrd\left(3, \{ \quad 2,,0 \quad \}, \cancel{OM}, RA\right)\right) = 4$$

2a

$$\{s,b\}$$
$$= (\{s,b\})\left\langle \begin{matrix} s \longleftarrow 2 \\ b \longleftarrow 0 \end{matrix} \right\rangle$$
$$= \{2, 0\}$$

2a.i.0

Tree

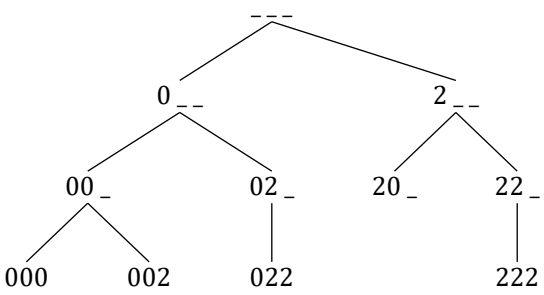


2a.i.1

$Wrd(3,\{2,0\},OM,RA) = \{000, 002, 020, 022, 200, 202, 220, 222\}$

2a.ii.0

Tree



2a.ii.1

$Wrd(3,\{2,0\},OM,RA) = \{000, 002, 022, 222\}$

6 (2b) Using a suitable tree (**that should be displayed on pages that follow, numbered as per instructions**) to **generate, count, and list** the elements of the set S of **distinct 4-digit** numbers that may be made by **using the elements of** $St\left(J \right)$ **as digits**:

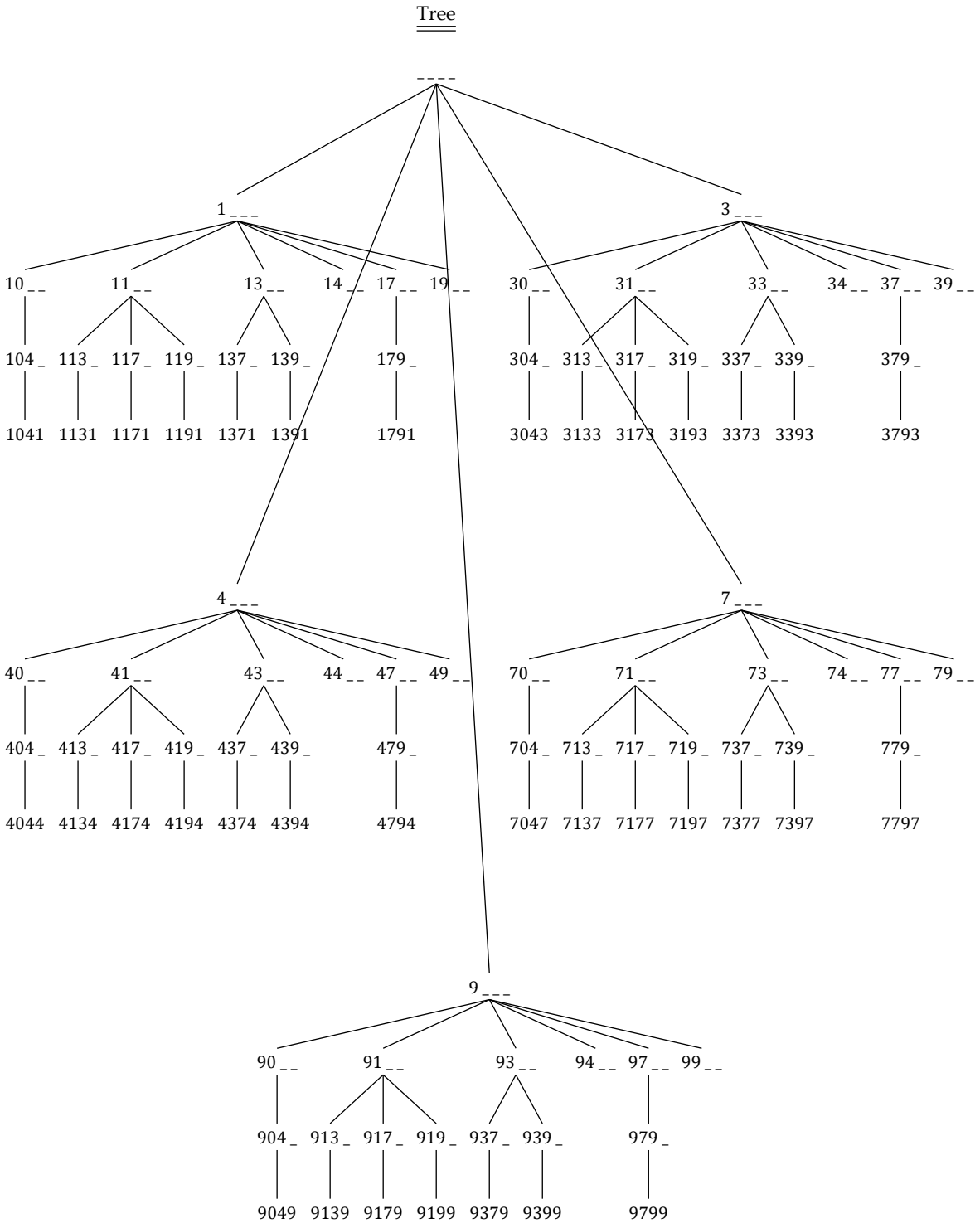
$$S = \left\{ \begin{array}{l} 1041, 1131, 1171, 1191, 1371, 1391, 1791, \\ 3043, 3133, 3173, 3193, 3373, 3393, 3793, \\ 4044, 4134, 4174, 4194, 4374, 4394, 4794, \\ 7047, 7137, 7177, 7197, 7377, 7397, 7797, \\ 9049, 9139, 9179, 9199, 9379, 9399, 9799 \end{array} \right\}$$

such that:

- 0 cannot be used in the first place
- the same digit occupies the first and fourth place,
- the third digit is strictly larger than the second digit, and
- the sum of the digits is even.

E The 4-digit number 2352 since 2 occupies the first and fourth positions, and 5 is strictly larger than 3 and the sum of the digits is 12, an even number. The 4-digit number 2342 does not qualify as the sum of the digits is 11, not an even number. The number 2532 does not qualify as 3 is not strictly larger than 5. The number 2345 does not qualify as two distinct numbers occupy the first and fourth place.

2b.0



2b.1

1041, 1131, 1171, 1191, 1371, 1391, 1791,
3043, 3133, 3173, 3193, 3373, 3393, 3793,
4044, 4134, 4174, 4194, 4374, 4394, 4794,
7047, 7137, 7177, 7197, 7377, 7397, 7797,
9049, 9139, 9179, 9199, 9379, 9399, 9799

5 (2c) Find a recursive definition for the function defined below, indicating the base case(s) and recursive step explicitly.

$$\rho \coloneqq \ n \longmapsto 2n(n+1) \ : \mathbb{N} \longrightarrow \mathbb{N}$$

Write your answers on this page in the areas provided.

1 (i) BC $\rho(0) \coloneqq 0$

4 (i) RcS $\forall n \in \mathbb{N}\ \rho(n+1) \coloneqq \rho(n) + 4(n+1)$

6 (3a) **Fill in the blanks appropriately** on this page and answer the following:

$s = 2$ $b = 0$ $b + s = 2$

1 (i) Compute, using the **binomial theorem**:

$$BnmlCfn\left(x^b y^s, (x+y)^{(b+s)}\right) = BnmlCfn\left(x^{(0\ )} y^{(2\ )}, (x+y)^{(2\ )}\right) = \binom{2}{2}$$

1 (ii) Compute, using the **binomial theorem**:

$$BnmlCfn\left(x^s y^b, (x+y)^{(s+b)}\right) = BnmlCfn\left(x^{(2\ )} y^{(0\ )}, (x+y)^{(2\ )}\right) = \binom{2}{0}$$

1 (iii) Compute, using the **multinomial theorem**:

$$MltnmlCfn\left(x^b y^s, (x+y)^{(b+s)}\right) = MltnmlCfn\left(x^{(0\ )} y^{(2\ )}, (x+y)^{(2\ )}\right) = \binom{2}{0\ 2}$$

1 (iv) Compute, using the **multinomial theorem**:

$$MltnmlCfn\left(x^s y^b, (x+y)^{(s+b)}\right) = MltnmlCfn\left(x^{(2\ )} y^{(0\ )}, (x+y)^{(2\ )}\right) = \binom{2}{2\ 0}$$

1 (v) The answers in (i), (ii), (iii) and (iv) are equal; record answer on this page and work on the facing page:

(Y) N (Pf) W

1 (vi) The answer to (v) depend on the values of s and of b; record answer on this page and work on the facing page:

Y (N) (Pf) W

3a

$$\frac{\frac{s=2 \; b=0 \; k=s \; n-k=b}{n-s=b}}{\frac{n=b+s}{\frac{n=0+2}{n=2}}}$$

3a.i

$$\begin{aligned}
& BnmlCfn(x^b\, y^s, (x+y)^{b+s}) \\
= \; & BnmlCfn(x^{2-2}\, y^2, (x+y)^2) \\
= \; & (BnmlCfn(a^{n-k} b^k, (a+b)^n)) \left\langle \begin{matrix} n \leftarrow 2 \\ k \leftarrow 2 \\ a \leftarrow x \\ b \leftarrow y \end{matrix} \right\rangle \\
= \; & \binom{n}{k} \left\langle \begin{matrix} n \leftarrow 2 \\ k \leftarrow 2 \end{matrix} \right\rangle \\
= \; & \binom{2}{2}
\end{aligned}$$

3a.ii

$$\begin{aligned}
& BnmlCfn(x^s\, y^b, (x+y)^{s+b}) \\
= \; & BnmlCfn(x^{2-0}\, y^0, (x+y)^2) \\
= \; & (BnmlCfn(a^{n-k} b^k, (a+b)^n)) \left\langle \begin{matrix} n \leftarrow 2 \\ k \leftarrow 0 \\ a \leftarrow x \\ b \leftarrow y \end{matrix} \right\rangle \\
= \; & \binom{n}{k} \left\langle \begin{matrix} n \leftarrow 2 \\ k \leftarrow 0 \end{matrix} \right\rangle \\
= \; & \binom{2}{0}
\end{aligned}$$

3a.iii

$$\begin{aligned}
& MltnmlCfn(x^b\, y^s, (x+y)^{b+s}) \\
= \; & (MltnmlCfn((x_1{}^{n_1})(x_2{}^{n_2}), (x_1+x_2)^n) \left\langle \begin{matrix} n \leftarrow 2 \\ n_1 \leftarrow 0 \\ n_2 \leftarrow 2 \\ x_1 \leftarrow x \\ x_2 \leftarrow y \end{matrix} \right\rangle \\
= \; & \binom{n}{n_1 \; n_2} \langle \text{"} \rangle \\
= \; & \binom{2}{0 \; 2}
\end{aligned}$$

3a.iv

$$MltnmlCfn(x^s\,y^b,(x+y)^{s+b})$$

$$= (MltnmlCfn((x_1{}^{n_1})(x_2{}^{n_2}),(x_1+x_2)^n)\left\langle \begin{matrix} n \longleftarrow 2 \\ n_1 \longleftarrow 2 \\ n_2 \longleftarrow 0 \\ x_1 \longleftarrow x \\ x_2 \longleftarrow y \end{matrix} \right\rangle$$

$$= \binom{n}{n_1\ n_2}\langle " \rangle$$

$$= \binom{2}{2\ 0}$$

3a.v.0

$$\binom{2}{2}$$

$$= \frac{(2!)}{((2!)((2-2)!))}$$

$$= \frac{(2!)}{((2!)(0!))}$$

3a.v.1

$$\binom{2}{0}$$

$$= \frac{(2!)}{((0!)((2-0)!))}$$

$$= \frac{(2!)}{((0!)(2!))}$$

3a.v.2

$$\binom{2}{0\ 2}$$

$$= \binom{2}{2\ 0}$$

$$= \frac{(2!)}{((0!)(2!))}$$

3a.v.3

$$\binom{2}{2}$$

$$= \binom{2}{0}$$

$$= \binom{2}{0\ 2}$$

$$= \binom{2}{2\ 0}$$

3a.vi.0

$$\begin{pmatrix} b+s \\ b \end{pmatrix}$$
$$= \frac{(b+s)!}{((b!)(((b+s))-b)!)}$$
$$= \frac{(b+s)!}{((b!)(s!))}$$

3a.vi.1

$$\begin{pmatrix} b+s \\ s \end{pmatrix}$$
$$= \frac{(b+s)!}{((s!)(((b+s))-s)!)}$$
$$= \frac{(b+s)!}{((s!)(b!))}$$

3a.vi.2

$$\begin{pmatrix} b+s \\ b\ \ s \end{pmatrix}$$
$$= \begin{pmatrix} b+s \\ s\ \ b \end{pmatrix}$$
$$= \frac{(b+s)!}{((b!)(s!))}$$

3a.vi.3

$$\begin{pmatrix} b+s \\ s \end{pmatrix}$$
$$= \begin{pmatrix} b+s \\ b \end{pmatrix}$$
$$= \begin{pmatrix} b+s \\ b\ \ s \end{pmatrix}$$
$$= \begin{pmatrix} b+s \\ s\ \ b \end{pmatrix}$$

6 (3b) Fill in the blank appropriately, and **show your work on pages that follow, numbered as per instructions**, to compute the value of, after **filling in the upper entry in the multinomial coefficient**:

$$\sum_{p+q=4} \binom{4}{0\ p\ q}$$

$$= \ 2+(3!)+(2)\left(\frac{4!}{3!}\right)$$

3b.0

$$\frac{\frac{\frac{\frac{n=n_1+n_2+n_3\left\langle \begin{matrix} n_1 \longleftarrow 0 \\ n_2 \longleftarrow p \end{matrix} \right\rangle}{n=0+p+q\left\langle\, n_3 \longleftarrow q \,\right\rangle}}{n=0+(p+q)}}{n=0+4\left\langle\, (p+q) \longleftarrow 4 \,\right\rangle}}{n=4}$$

3b.1

$$\begin{aligned} & 4 \\ = \; & 0+0+4 \\ = \; & 0+1+3 \\ = \; & 0+2+2 \\ = \; & 0+3+1 \\ = \; & 0+4+0 \end{aligned}$$

3b.2

$$\sum_{p+q=4} \binom{4}{p \;\; q}$$

$$\left| \begin{aligned} = & \binom{4}{0 \;\; 0 \;\; 4} \\ + & \binom{4}{0 \;\; 1 \;\; 3} \\ + & \binom{4}{0 \;\; 2 \;\; 2} \\ + & \binom{4}{0 \;\; 3 \;\; 1} \\ + & \binom{4}{0 \;\; 4 \;\; 0} \end{aligned} \right| \left| \begin{aligned} = & \; 1 \\ + & \; \frac{4!}{3!} \\ + & \; 3! \\ + & \; \frac{4!}{3!} \\ + & \; 1 \end{aligned} \right| = 2+(3!)+(2)\left(\frac{4!}{3!}\right)$$

3b.3

$$\begin{aligned} & \binom{4}{0 \;\; 0 \;\; 4} \\ = \; & \binom{4}{0 \;\; 4 \;\; 0} \\ = \; & \frac{4!}{(0!)(0!)(4!)} \\ = \; & \frac{4!}{(1)(1)(4!)} \\ = \; & \frac{4!}{4!} \\ = \; & 1 \end{aligned}$$

3b.4

$$\begin{aligned}
&\binom{4}{0\ 1\ 3} \\
&= \binom{4}{0\ 3\ 1} \\
&= \frac{4!}{(0!)(3!)(1!)} \\
&= \frac{4!}{(1)(3!)(1)} \\
&= \frac{4!}{3!}
\end{aligned}$$

3b.5

$$\begin{aligned}
&\binom{4}{0\ 2\ 2} \\
&= \frac{4!}{(0!)(2!)(2!)} \\
&= \frac{4!}{(1)(2)(2)} \\
&= 3!
\end{aligned}$$

5 (3c) Prove by induction that: $\forall n \in \mathbb{N}\left(4|\left(\rho(n)\right)\right)$ using your recursive definition from (2c) for the function defined below:

$$\rho \coloneqq \ n \longmapsto 2n(n+1) \ : \mathbb{N} \longrightarrow \mathbb{N}$$

Begin your work on this page and continue onto pages that follow, numbered as per instructions, as needed.

SPf
$\forall n \in \mathbb{N}\ P(n) : \Leftrightarrow (4|\rho(n))$

0 (i) BC

$\rho(0)=0$
$\overline{\rho(0)=(0)(4)}$
$\overline{P(0)}$

0 (ii) RcS

$P(0)\ P(1)\ \ldots\ P(n)$
$\overline{\rho(n)=4c}$

$\rho(n+1)$
$= \ \rho(n)+4(n+1)$
$= \ 4c+4(n+1)$
$= \ (c+(n+1))(4)$
$\overline{P(n+1)}$

Hence $\dfrac{P(0)\ (\ \dfrac{P(0)\ P(1)\ \ldots\ P(n)}{P(n+1)}\ )}{\forall n \in \mathbb{N}(P(n))}$

EPf